Atlas 210X — system wiring of the modifications (YT3CT, 2026-10)

Block level. Component details: wiring_A4 section numbers in [ ]. Diode band = cathode. The Nano never sees more than 5 V: diodes / dividers at the source end.

FRONT PANEL

LCD RG1602A 16x2 [1]
12 wires, twisted groups, 470 Ω at the Nano
contrast pot on the Nano board (V0 yellow)

Tuning encoder MBL600
A / B, 100 detents, own decoder

STEP + MENU buttons, MODE rocker [3]
STEP 0 Ω, MENU 2.2 k, MODE (OPP) 22 k
A6 ladder, 100 nF at the buttons

ATT / NORM / PRE rocker [4]
SPDT ON-OFF-ON, common = GND
throws to the relay coil low ends

AGC fast / slow rocker [5]
DPDT: pole 1 GND -> BC547 base (FAST),
pole 2 GND -> D4 (no divider)

AF / RF gain (concentric) [7]
AF 10 k: VariBeam out -> PC-300 pin 12
RF 5 k: R5 2.2 k, R7 220 Ω -> PC-100 pin 14

MIC / ALC gain (concentric) [7]
MIC 20 k (20 k across), R13 10 k -> +CW
ALC 100 k (11 k across)

VariBeam control (own Nano)
knob D2/D3/D4, OLED SSD1306, D5/D6/D7 -> VB
D8 <- VFO D13 (CAT PTT, 300 ms filter)
D9 -> 4.7 k -> BC547 -> PTT relay coil

Mic jack (front)
Shure 404C, PTT on the KEY line, mic -> relay NC

ATU-100 buttons
TUNE / AUTO / BYPASS -> ATU-100 only

Function switch S3
OFF-CAL-REC-TRANS-CW (original)
CW = TX + +CW, CAL = cal line

LCD 12 wires (5 twisted groups + V0) -> D7-D12
encoder A / B -> D2 / D3
A6 + GND (twisted, green) -> A6
ATT / PRE coil low ends -> D5 / D6
D4 orange (AGC pole 2) -> D4
D13 -> 10 k -> VariBeam Nano D8 -> D13
serial (twisted 4) -> D0 / D1 / RST

VFO NANO

D7-D12
A4 / A5 (I2C)
D2 / D3
A6
A0 / A1
D5 / D6
A3
A2
D4
+5V / GND
A7
D13
D0 / D1 / RST

sketch atlas_vfo_fixed

I2C SDA / SCL -> **Si5351 module (25 MHz)**
PLLA -> CLK0 LO, even divider (spur-free choice)
PLLB -> CLK1 BFO 5.520 MHz (+800 Hz CW,
+3.3 kHz MODE OPP)
CLK2 CAL marker / AUTO CAL 2.5 MHz

LPF CLK0
LPF CLK1

A7 network at S3 [6]
+CW 1N4148 22 k, CAL 1N4148 47 k,
10 k to GND (one blue wire to A7)

J4 rear socket (old EXT OSC) [8] [10]
1 RST (1 µF in series, from the adapter DTR)
2 TX green 3 RX white 4 GND black
5 audio in 6 audio GND (floating) 7 audio out
8 / 9 spare (PTT = CAT command)
USB-TTL adapter outside: CAT + uploads
one USB cable for CAT, PTT, audio separate

PTT relay DPDT [10]
BC547 <- VariBeam D9

T1 / T2 600:600

CLK2 thin coax [9]
BFO (CLK1) coax
LO (CLK0) coax
A0 / A1 S-meter: 10 k + 100 nF each, 1 k across 15-16
A3 band switch ladder
A2 white, 1N4148 at the Nano (band to KEY line)
+CW (C6 / R14) and CAL line
D5 grey / D6 red (1N4148 at the coils)
AUTO CAL: D5 <- CAL box tap (plain wire), D6 <- GPSDO 1PPS via 1 k (removable)
KEY line + mic hot (NC)
AGC pole 1: GND in FAST -> BC547 base
AF / RF / MIC / ALC pot wiring (original wires) [7]
VariBeam Nano D5/D6/D7 -> DSP board (emulated encoder)

ATLAS 210X BOARDS / CHASSIS

PC-100 (1st mixer / IF / relay)
LO in from CLK0 | pins 15 / 16 S-meter bridge
pin 14 RF gain | pin 20 KEY line (LOW = TX)
pin 18 WHT/BLK +CW

BFO / carrier input
original PC-500 output point (PC-500 removed)

S4 band switch
resistor ladder on the band wafer -> A3

KEY line
A2 diode, PTT relay (pole 1), mic PTT, S3

PC-200
pin 17 mic in <- J4 audio in (47 k / 470 Ω)
pin 16 mic gain top <- R13 10 k from +CW
pin 19 AGC | CA3086 mic / S-meter amp

PC-300D
pin 22 AF out -> VariBeam DSP + J4 audio out
pin 12 AF in <- AF gain wiper
AGC fast: BC547 switches C311 22 µF (2.2 µF stays)
R309 AGC threshold 5 k multi-turn

VariBeam DSP board
pin 22 -> DSP -> AF gain pot -> pin 12

S3 function switch / C6-R14
+CW 0 V REC/SSB, +13 V CW; brown = GND

CAL attenuator box (PC-820 removed) [9]
1 k / 2 x 10 Ω / 4.7 k -> ATT/PRE J1, header = AUTO CAL tap

ATT / PRE board (PC-800C OUT -> PC-100B pin 1) [4]
ATT ~17 dB π-pad, PRE BFG135 ~+8 dB, relays on the rocker
coil high end +12 V, low end to the rocker
PRE supply: BD140 + 78L08, only in PRE
RX line only, never in the TX path

Pots wiring / mic jack / ATU-100 (separate)